

Parallel Computing

Interconnection Networks

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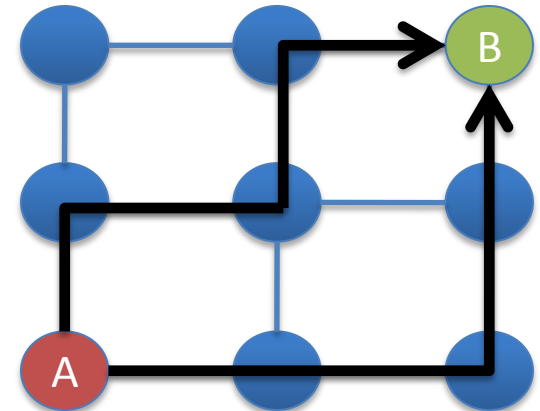
- Introduction and Terminology
- Topology
- Routing and Flow control

Interconnection Network Basics

- **Topology**
 - Specifies the way switches are wired
 - Affects routing, reliability, throughput, latency, building ease
- **Routing**
 - How does a message get from source to destination
 - Static or adaptive
- **Buffering and Flow Control**
 - What do we store within the network?
 - Entire packets, parts of packets, etc?
 - How do we manage and negotiate buffer space?
 - How do we throttle during oversubscription?
 - Tightly coupled with routing strategy

Topology & Routing

- **Topology:** Determines arrangement of nodes and links in network
- Significant impact on network performance
 - Determines number of **hops**
- **Routing:** Routing algorithm determines path(s) from source to destination

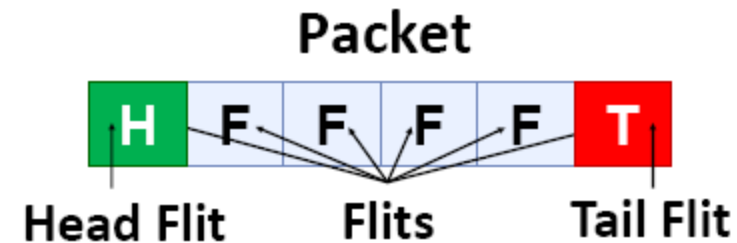


Terminology

- **Network interface**
 - Connects endpoints (e.g. cores) to network.
 - Decouples computation/communication
- **Links**
 - Bundle of wires that carries a signal
- **Switch/router**
 - Connects fixed number of input channels to fixed number of output channels
- **Channel**
 - A single logical connection between routers/switches

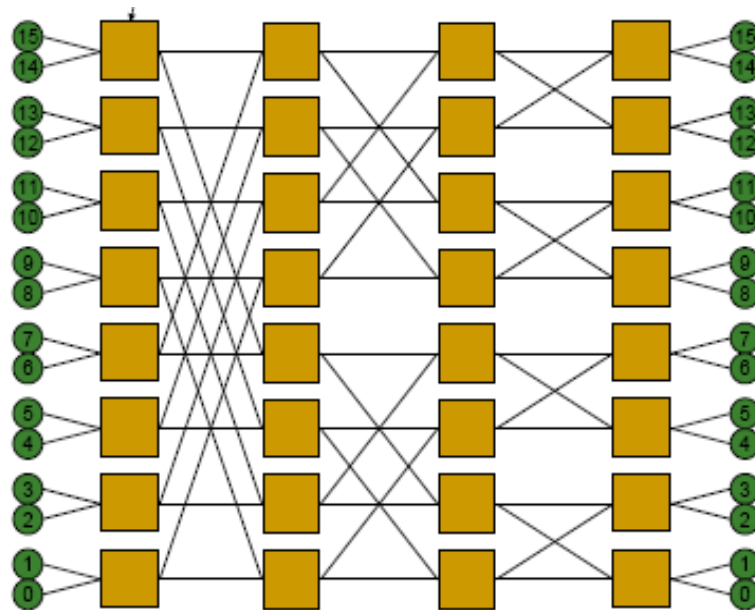
More Terminology

- **Node**
 - A network endpoint connected to a router/switch
- **Message**
 - Unit of transfer for network clients (e.g. cores, memory)
- **Packet**
 - Unit of transfer for network
- **Flit**
 - Flow control digit
 - Unit of flow control within network

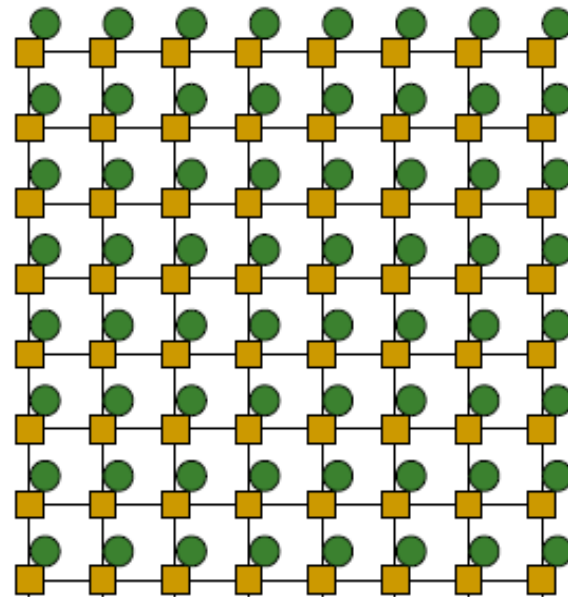


Some More Terminology

- **Direct or Indirect Networks**
 - Endpoints sit “inside” (direct) or “outside” (indirect) the network
 - E.g. mesh is direct; every node is both endpoint and switch
 - Router (switch), Radix of 2 (2 inputs, 2 outputs) Abbreviation: Radix-ary
These routers are 2-ary



Indirect



Direct

Properties of a Topology/Network

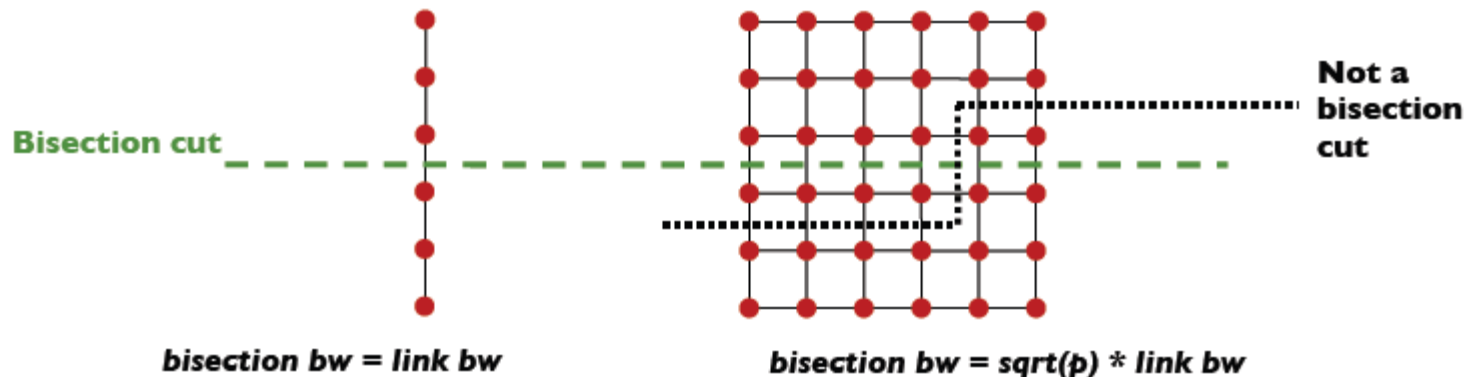
- **Regular or Irregular**
 - regular if topology is regular graph (e.g. ring, mesh)
- **Routing Distance**
 - number of links/hops along route
- **Diameter**
 - maximum routing distance
- **Average Distance**
 - average number of hops across all valid routes

Properties of a Topology/Network

- **Blocking vs. Non-Blocking**
 - If connecting any permutation of sources & destinations is possible, network is non-blocking; otherwise network is blocking.
- **Bisection Bandwidth**
 - Often used to describe network performance
 - Cut network in half and sum bandwidth of links severed
 - (Min # channels spanning two halves) * (BW of each channel)
 - Meaningful only for recursive topologies
 - Can be misleading, because does not account for switch and routing efficiency

Bisection Bandwidth

- Definition: # links across smallest cut that divides nodes in two (nearly) equal parts
- Important for **all-to-all communication**
- Variation: Bisection *bandwidth* = *bandwidth across smallest cut*

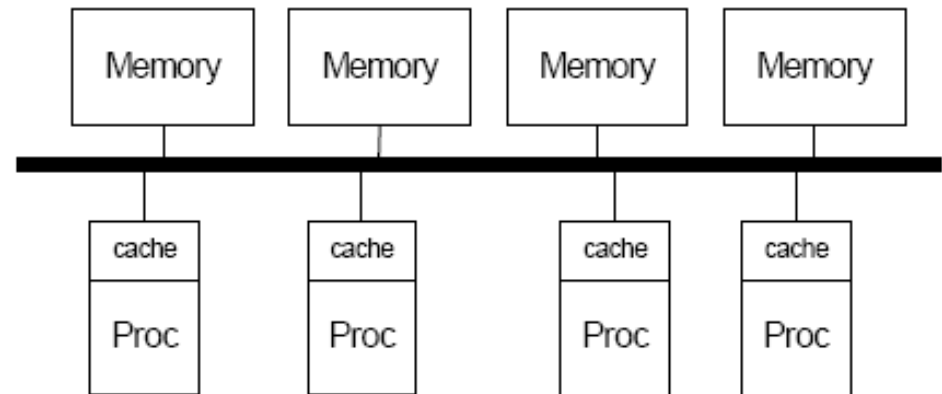


Many Topology Examples

- Bus
- Crossbar
- Ring
- Tree
- Omega
- Hypercube
- Mesh
- Torus
- Butterfly
- ...

Bus

- + Simple
- + Cost effective for a small number of nodes
- + Easy to implement coherence (snooping)
- Not scalable to large number of nodes
(limited bandwidth, electrical loading -> reduced frequency)
- High contention

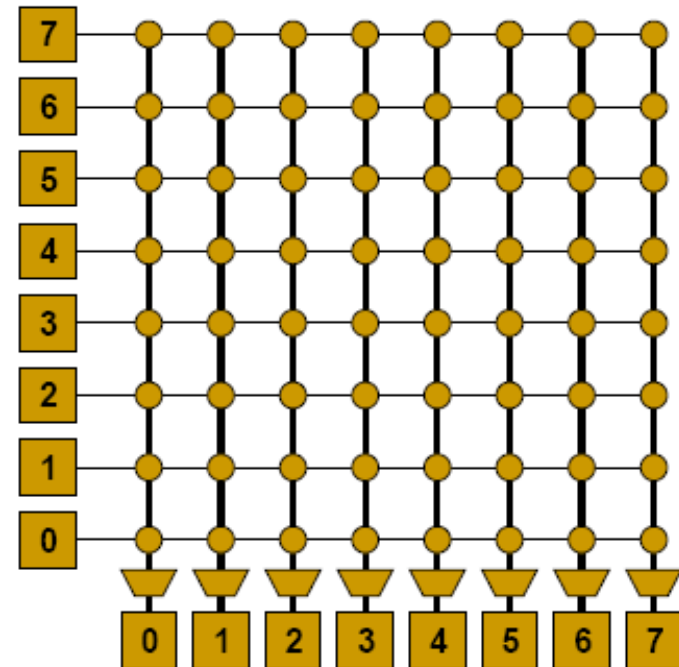


Crossbar

- Every node connected to all others (non-blocking)
- Good for small number of nodes
 - + Low latency and high throughput
 - Expensive
 - Not scalable -> $O(N^2)$ cost
 - Difficult to arbitrate

Core-to-cache-bank networks:

- IBM POWER5
- Sun Niagara I/II

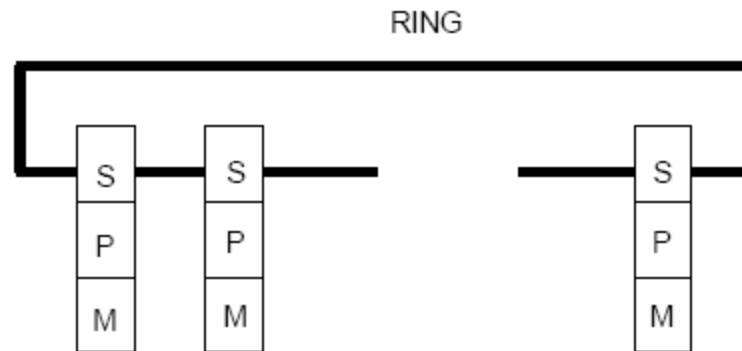


Ring

- + Cheap: $O(N)$ cost
- High latency: $O(N)$
- Not easy to scale
- Bisection bandwidth remains constant

Used in:

- Intel Larrabee/Core i7
- IBM Cell



Linear and ring networks

- **Diameter** : Length of shortest path between farthest pair
- **Bisection bandwidth** : bandwidth across smallest cut that bisects network
- **Average distance**

Linear: $P-1$ links

Diameter = ?

Avg. dist. = ?

Bisection = ?



Ring/Torus: P links

Diameter = ?

Avg. dist. = ?

Bisection = ?



Linear and ring networks

Linear: $P-1$ links

Diameter = $P-1$

Avg. dist. $\sim P/3$

Bisection = 1



Ring/Torus: P links

Diameter $\sim P/2$

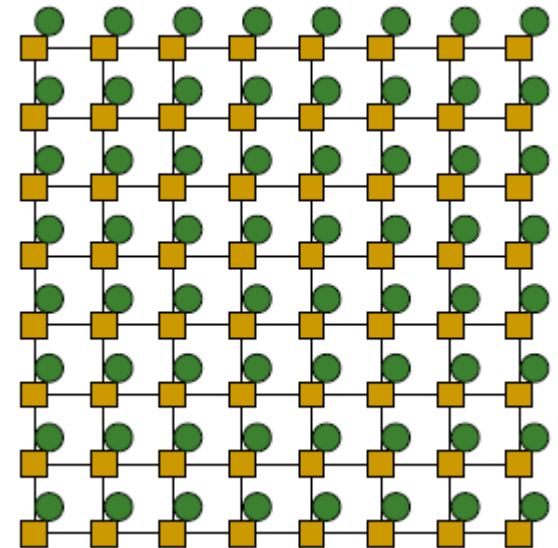
Avg. dist. $\sim P/4$

Bisection = 2



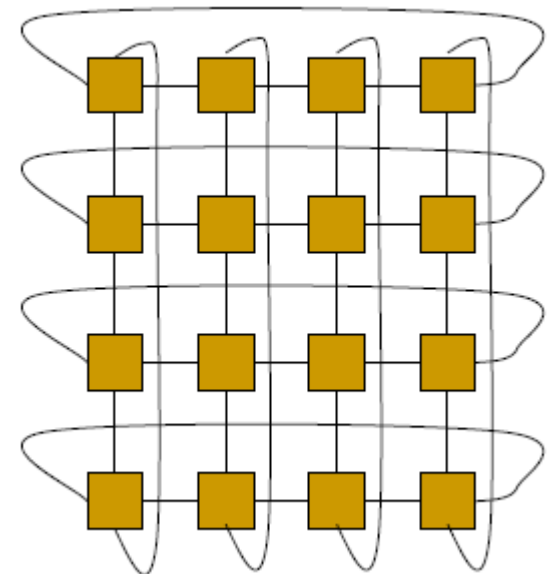
Mesh

- $O(N)$ cost
- Average latency: $O(\sqrt{N})$
- Easy to layout on-chip: regular & equal-length links
- Path diversity: many ways to get from one node to another
- Used in:
 - Tiler 100-core CMP
 - On-chip network prototypes



Torus

- Mesh is not symmetric on edges:
performance very sensitive to placement of
task on edge vs. middle
- Torus avoids this problem
 - + Higher path diversity (& bisection bandwidth)
than mesh
 - Higher cost
 - Harder to lay out on-chip
 - Unequal link lengths



Trees

Planar, hierarchical topology

Latency: $O(\log N)$

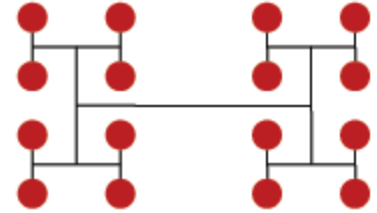
Good for local traffic

+ Cheap: $O(N)$ cost

+ Easy to Layout

- Root can become a bottleneck

Fat trees avoid this problem (CM-5)



Hypercube

- Latency: $O(\log N)$
- Radix: $O(\log N)$
- #links: $O(N \log N)$
- + Low latency
- - Hard to lay out in 2D/3D
- Used in some early message passing machines, e.g.:
 - Intel iPSC
 - nCube

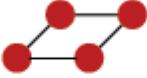
No. of nodes = 2^d for dimension d

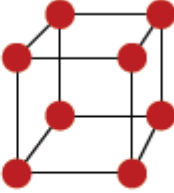
◦ Diameter = d

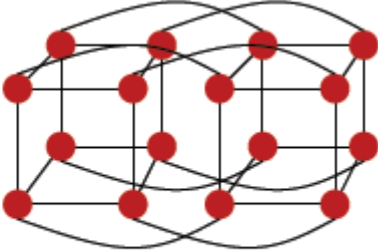
◦ Bisection = $p/2$


 $d=0$


1

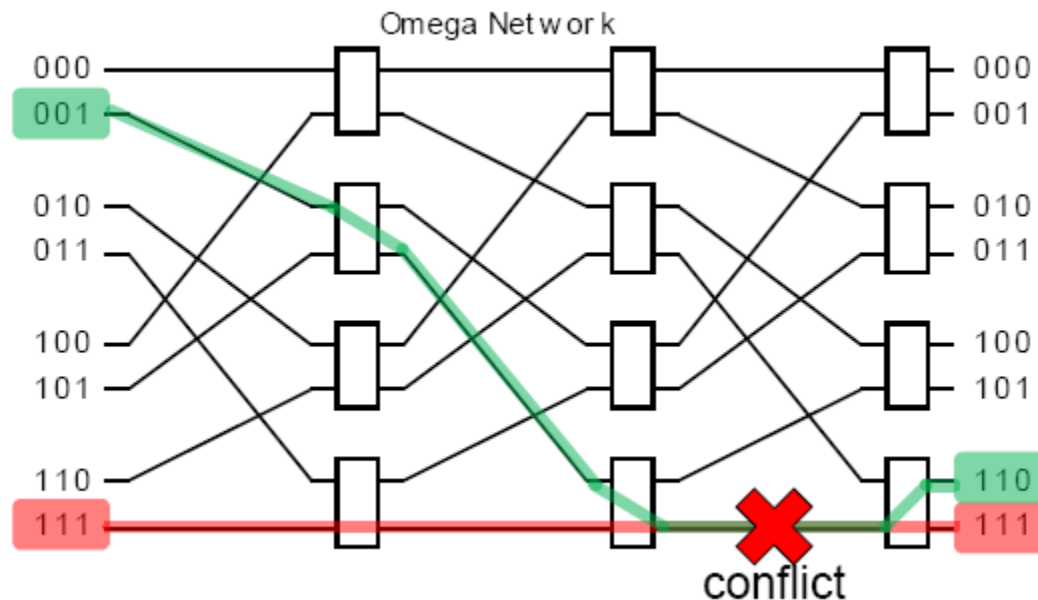

2


3


4

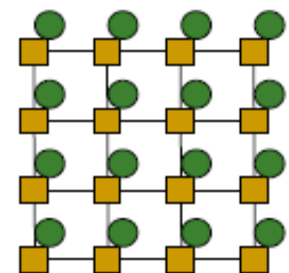
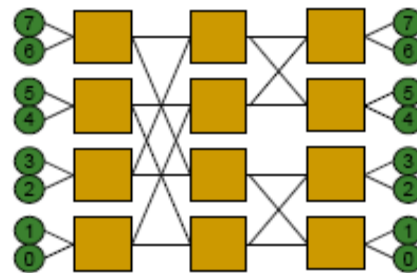
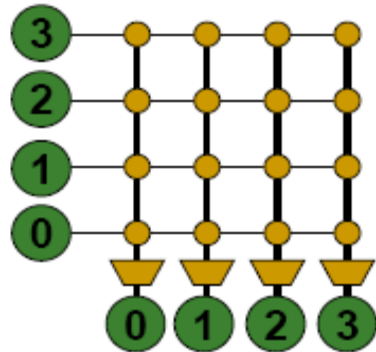
Multistage Logarithmic Networks

- Idea: Indirect networks with multiple layers of switches between terminals
- Cost: $O(N \log N)$, Latency: $O(\log N)$
- Many variations (Omega, Butterfly, Benes, Banyan, ...)
- E.g. Omega Network:



Q: Blocking or non-blocking?

Review: Topologies



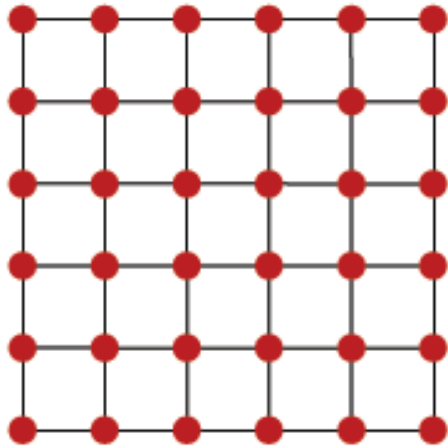
Topology	Crossbar	Multistage Logarith.	Mesh
Direct/Indirect	Indirect	Indirect	Direct
Blocking/ Non-blocking	Non-blocking	Blocking	Blocking
Cost	$O(N^2)$	$O(N \log N)$	$O(N)$
Latency	$O(1)$	$O(\log N)$	$O(\sqrt{N})$

Multidimensional meshes and tori

2-D mesh: $\sim 2 \cdot P$ links

Diameter = ?

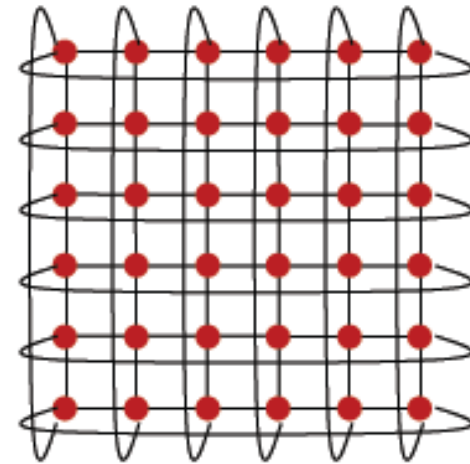
Bisection = ?



2-D torus: $2 \cdot P$ links

Diameter = ?

Bisection = ?

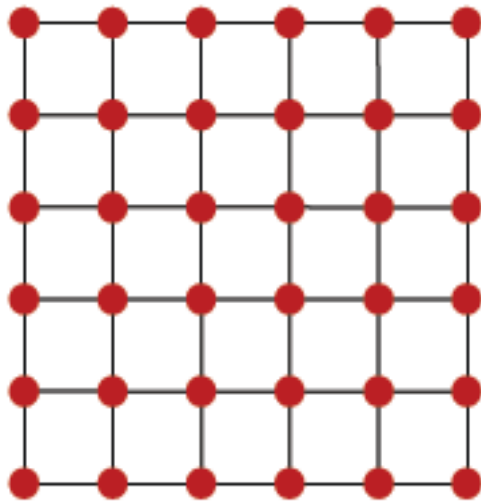


Multidimensional meshes and tori

2-D mesh: $\sim 2 \cdot P$ links

Diameter $\sim 2 \cdot \sqrt{P}$

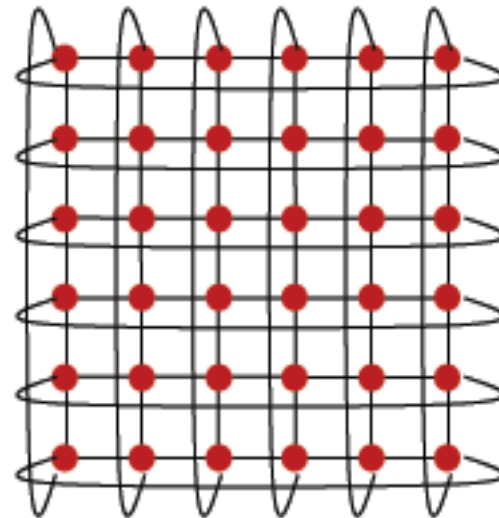
Bisection = $\sqrt{P}$



2-D torus: $2 \cdot P$ links

Diameter $\sim \sqrt{P}$

Bisection = $2 \cdot \sqrt{P}$

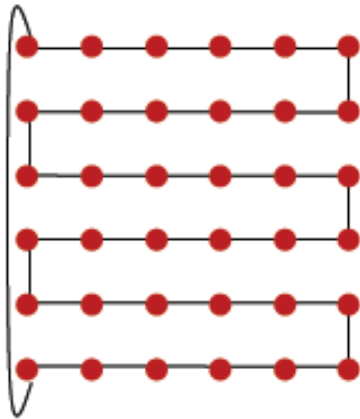


Mappings and congestion

Ring: P links

Diameter $\sim P / 2$

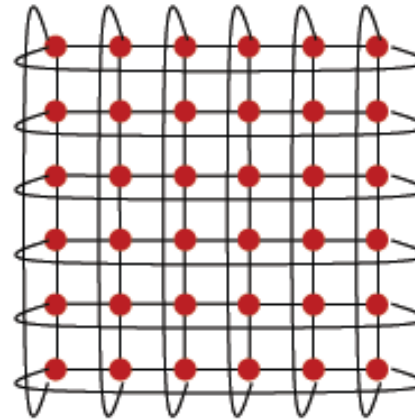
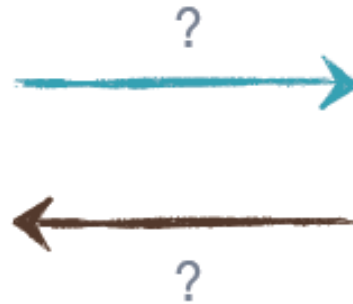
Bisection = 2



2-D torus: $2 \cdot P$ links

Diameter $\sim \sqrt{P}$

Bisection = $2 \cdot \sqrt{P}$



Node mapping implies an edge mapping.

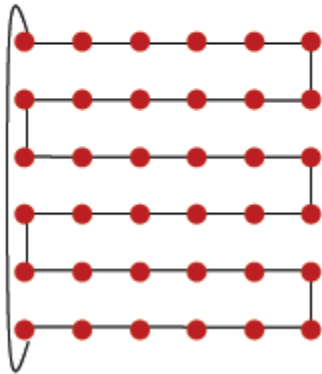
Congestion = maximum number of source edges that map to a target edge.

Mappings and congestion

Ring: P links

Diameter $\sim P / 2$

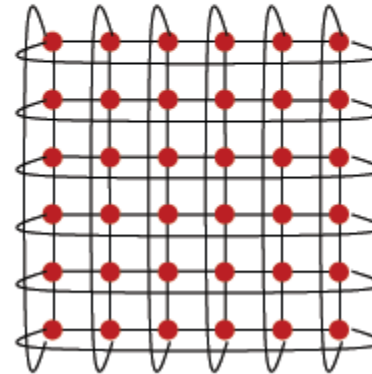
Bisection = 2



2-D torus: $2 \cdot P$ links

Diameter $\sim \sqrt{P}$

Bisection = $2 \cdot \sqrt{P}$



$C = 1$

?

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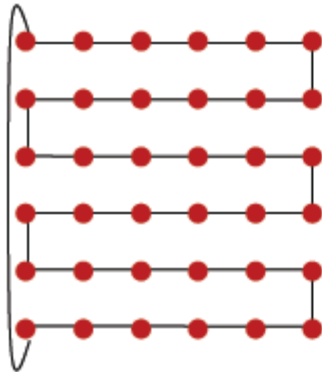
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Mappings and congestion

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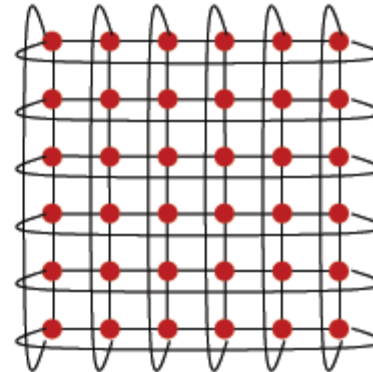
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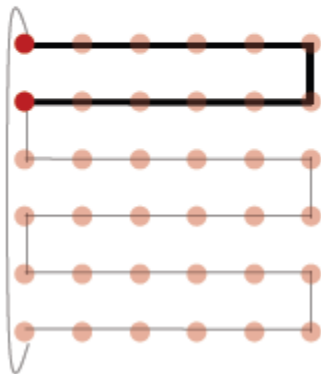
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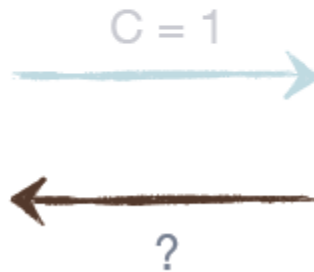
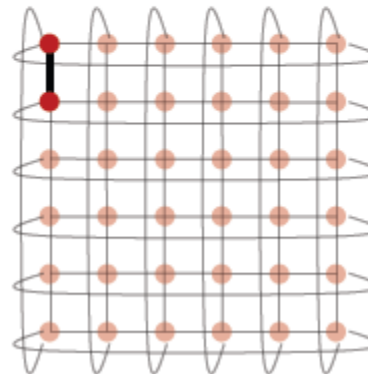
Bisection = 2



2-D torus: $2 \cdot P$ links

Diameter $\sim \sqrt{P}$

Bisection = $2 \cdot \sqrt{P}$



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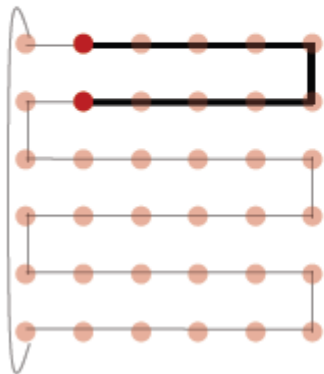
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Mappings and congestion

Ring: P links

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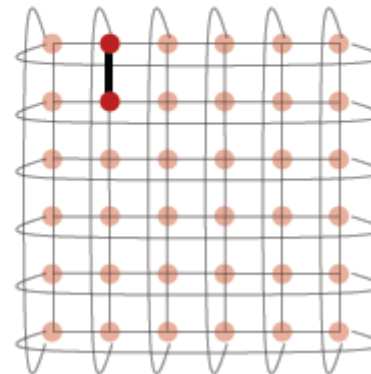
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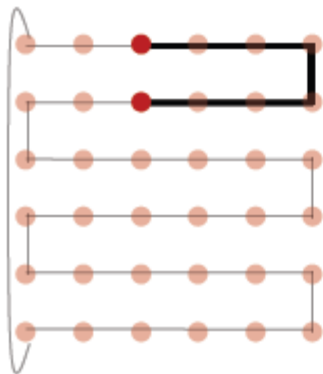
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Mappings and congestion

Ring: P links

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Bisection = 2



$C = 1$

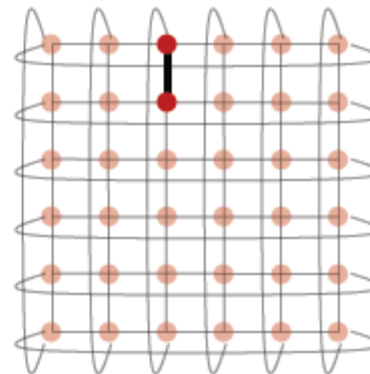


?

2-D torus: $2 \cdot P$ links

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Node mapping implies an edge mapping.

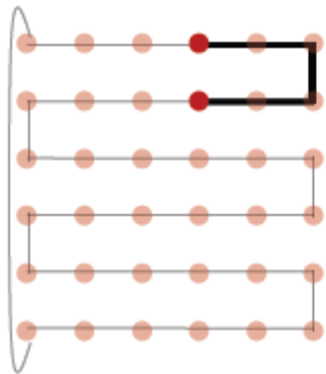
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Mappings and congestion

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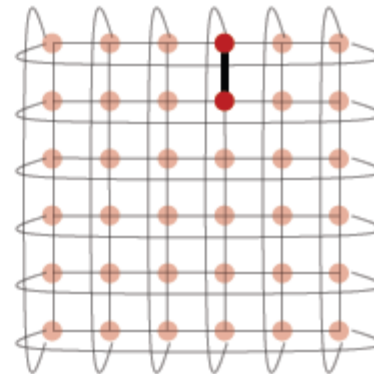
Bisection = 2



2-D torus: $2 \cdot P$ links

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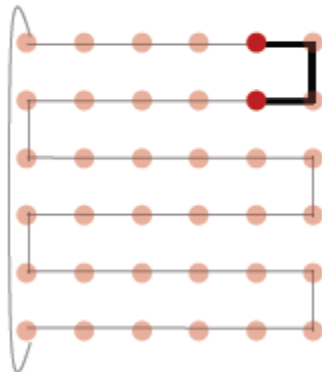
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Mappings and congestion

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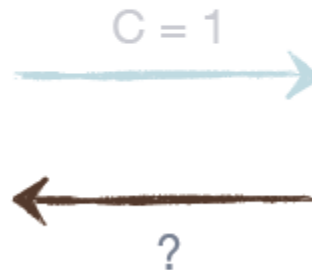
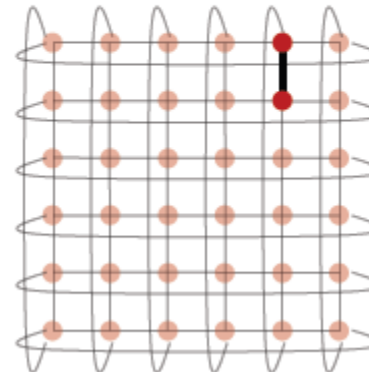
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2-D torus: $2 \cdot P$ links

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Node mapping implies an edge mapping.

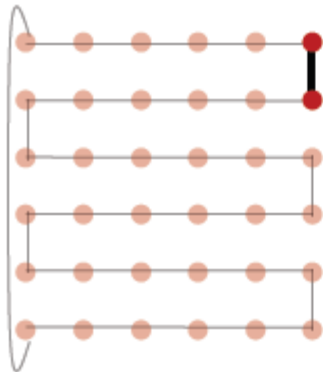
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Mappings and congestion

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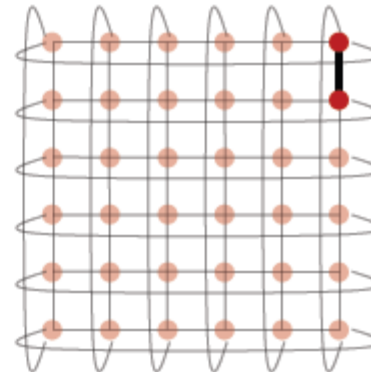
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2-D torus: $2 \cdot P$ links

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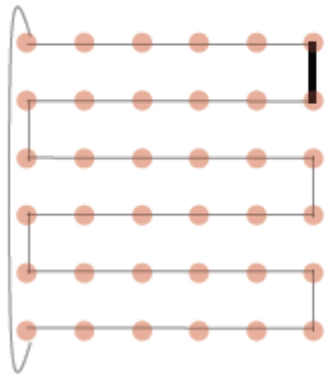
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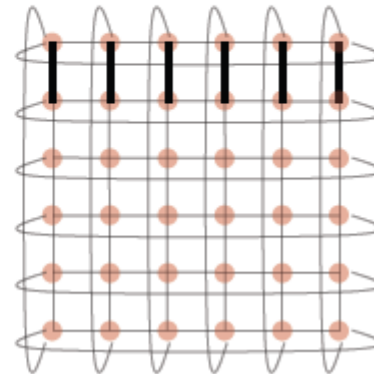
Bisection = 2



2-D torus: $2 \cdot P$ links

Diameter $\sim \sqrt{P}$

Bisection = $2 \cdot \sqrt{P}$



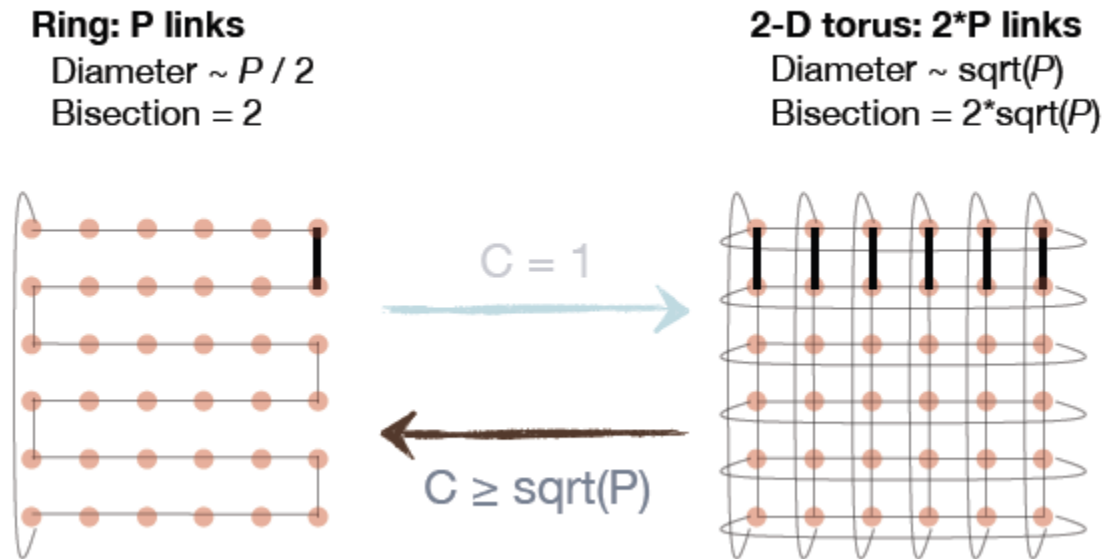
$C = 1$

$C \geq \sqrt{P}$

Node mapping implies an edge mapping.

Congestion = maximum number of source edges that map to a target edge.

Mappings and congestion



General principle:
Ratio of bisection widths is a **lower bound** on congestion

Node mapping implies an edge mapping.

Congestion = maximum number of source edges that map to a target edge.

Topology properties (*n nodes total*)

Topology	Diameter	Bisection	Arc connectivity	# links
Linear	$n - 1$	1	1	$n - 1$
Ring	$\approx n/2$	2	2	n
2-D mesh	$\approx 2\sqrt{n}$	$\sqrt{n}$	2	$n - 1$
2-D torus	$\approx \sqrt{n}$	$2\sqrt{n}$	4	$2n$
Hypercube	$\log n$	$n/2$	$\log n$	$1/2 \cdot n \log n$
k -ary tree	$2 \log_k n$	1	1	$n - 1$
Butterfly	$\log n$	n	n	$\approx n \log n$
d-D torus	$\approx \sqrt[d]{n} \cdot d/2$	$2n^{(d-1)/d}$	1	$n - 1$
Completely connected	1	$n^2/4$	$n - 1$	$n(n - 1)/2$

Source: Grama, et al. (2003), *Intro. to Parallel Computing*.

Topologies in practice

Machine	Network
ORNL Titan (Cray XK7)	3D torus
IBM Blue Gene/Q	5D torus
K computer	6D torus
Tianhe-1A (GPU)	Fat tree (?)
Tsubame (GPU)	Fat tree
Cray XE6	3D torus
Cray XT3, XT4, XT5	3D torus
BG/L, BG/P	3D torus (+ others)
SGI Altix	Fat tree
Cray X1	4D hypercube*
Millennium (UCB, Myricom)	Arbitrary*
HP Alphaserver (Quadrics)	Fat tree
IBM SP	~ Fat tree
SGI Origin	Hypercube
Intel Paragon	2D mesh
BBN Butterfly	Butterfly

“ α - β ” (latency-bandwidth) cost model

- Model time to send a message in terms of latency and bandwidth

$$t(n) = \alpha + \frac{n}{\beta}$$

α (latency) , β (bandwidth)

- Node may send to any other
- May send and receive Simultaneously
- Usually, $\text{cost}(\text{flop}) \ll 1/\beta \ll \alpha$
 - One long message cheaper than many short ones
 - Can do $\sim$ thousands of flops for each message
 - Want large computation-to communication ratio

Does network topology matter?

- Mapping algorithms to networks used to be a “hot topic”
 - Key metric: Minimize hops
 - Modern networks hide hop cost (e.g., wormhole routing) and software overheads dominate wire latencies, so topology seemed less important over time
- Gap in **hardware/software latency: On IBM SP, *cf. 1.5 usec to 36 usec***
- Topology affects **bisection bandwidth, so still relevant**

Wormhole flow control

Switching/Flow Control Overview

- Topology: determines connectivity of network
- Routing: determines paths through network
- Flow Control: determine allocation of resources to messages as they traverse network
 - Buffers and links
 - Significant impact on throughput and latency of network

Packets

- Messages: composed of one or more packets
 - If message size is $\leq$ maximum packet size only one packet created
- Packets: composed of one or more flits
- Flit: flow control digit
- Phit: physical digit
 - Subdivides flit into chunks = to link width
 - In on-chip networks, flit size == phit size.
 - Due to very wide on-chip channels

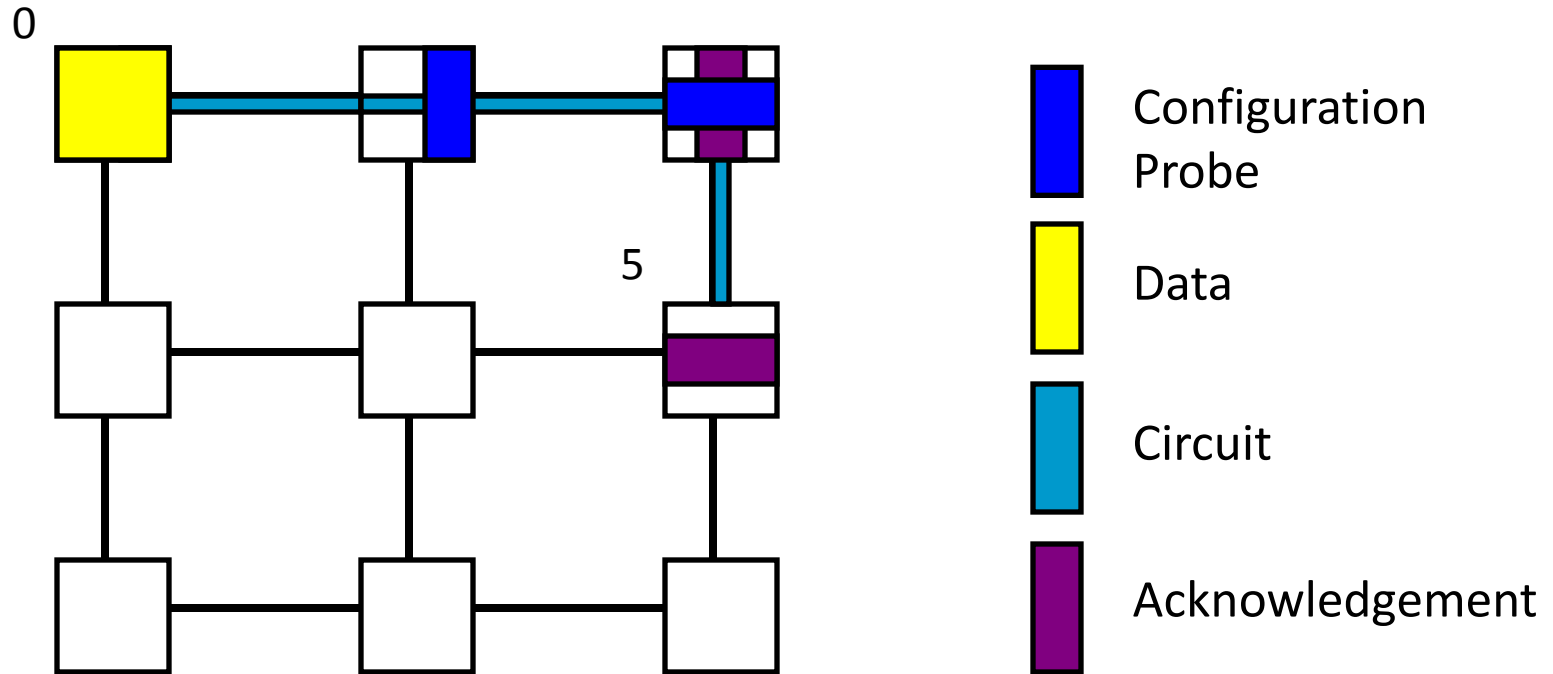
Switching

- Different flow control techniques based on granularity
- Circuit-switching: operates at the granularity of messages
- Packet-based: allocation made to whole packets
- Flit-based: allocation made on a flit-by-flit basis

Circuit Switching

- All resources (from source to destination) are allocated to the message prior to transport
 - Probe sent into network to reserve resources
- Once probe sets up circuit
 - Message does not need to perform any routing or allocation at each network hop
 - Good for transferring large amounts of data
 - Can amortize circuit setup cost by sending data with very low per-hop overheads
- No other message can use those resources until transfer is complete
 - Throughput can suffer due setup and hold time for circuits

Circuit Switching Example

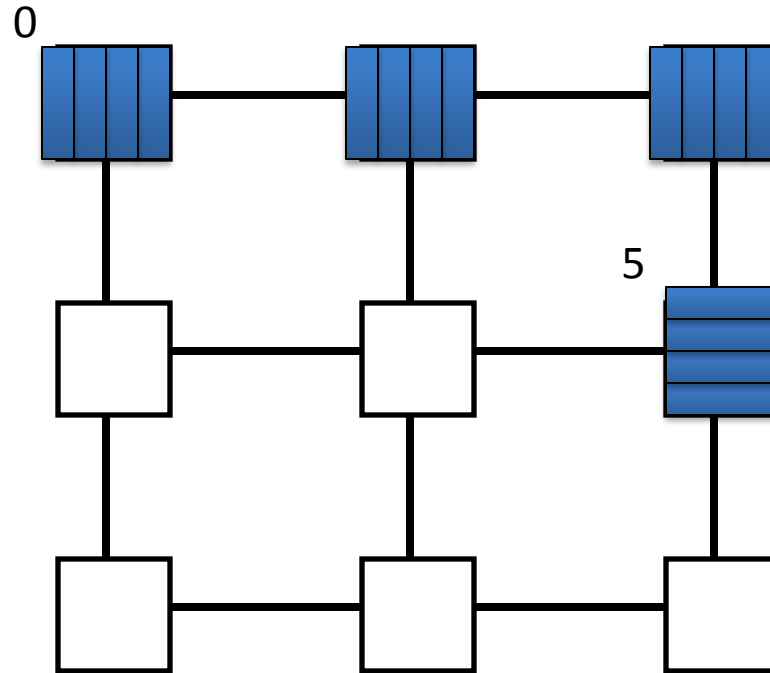


- Significant latency overhead prior to data transfer
- Other requests forced to wait for resources

Packet-based Flow Control

- Store and forward
- Links and buffers are allocated to entire packet
- Head flit waits at router until entire packet is buffered before being forwarded to the next hop
- Not suitable for on-chip
 - Requires buffering at each router to hold entire packet
 - Incurs high latencies (pays serialization latency at each hop)

Store and Forward Example

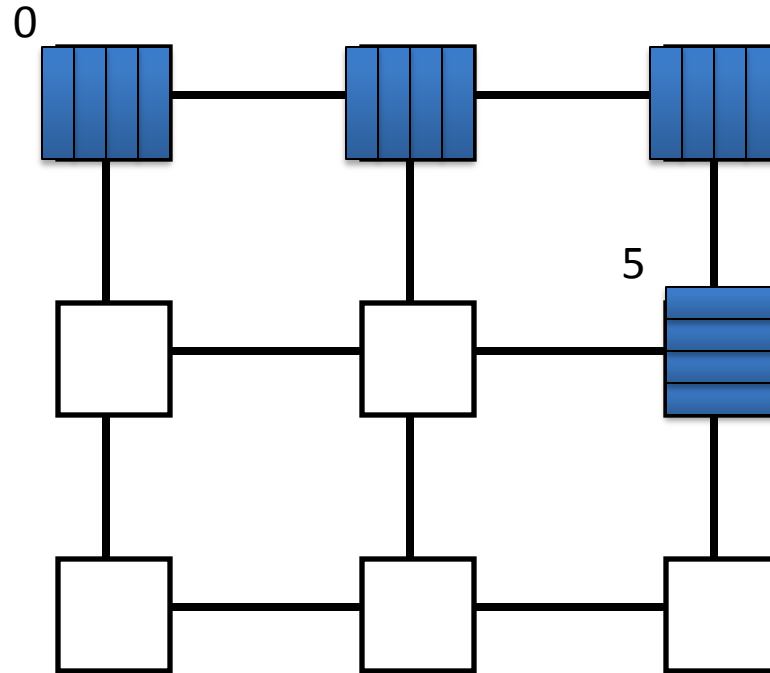


- High per-hop latency
- Larger buffering required

Virtual Cut Through

- Packet-based: similar to Store and Forward
- Links and Buffers allocated to entire packets
- Flits can proceed to next hop before tail flit has been received by current router
 - But only if next router has enough buffer space for entire packet
- Reduces the latency significantly compared to SAF
- But still requires large buffers
 - Unsuitable for on-chip

Virtual Cut Through Example

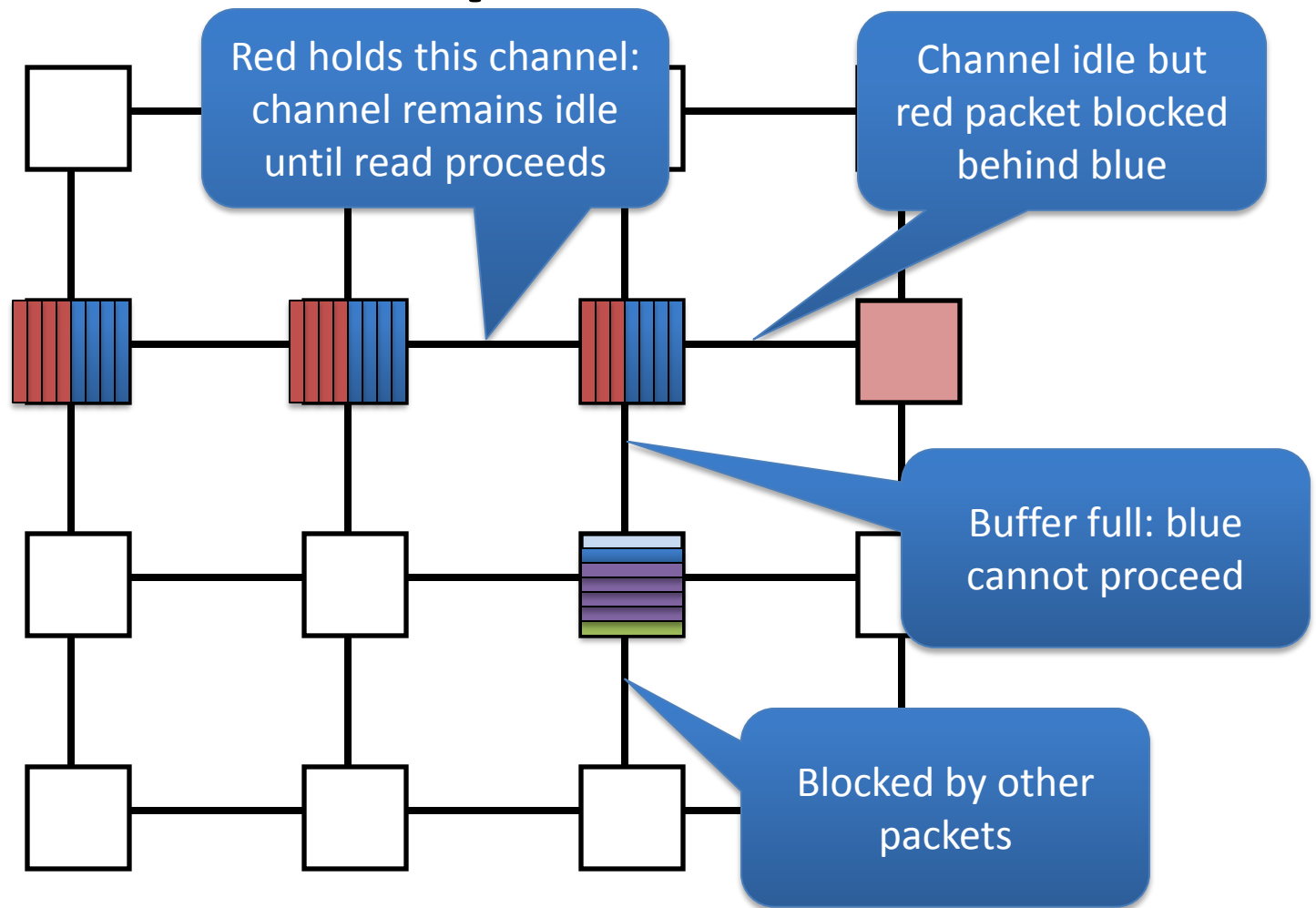


- Lower per-hop latency
- Larger buffering required

Flit Level Flow Control

- Wormhole flow control
- Flit can proceed to next router when there is buffer space available for that flit
 - Improved over SAF and VCT by allocating buffers on a flit-basis
- Pros
 - More efficient buffer utilization (good for on-chip)
 - Low latency
- Cons
 - Poor link utilization: if head flit becomes blocked, all links spanning length of packet are idle
 - Cannot be re-allocated to different packet
 - Suffers from head of line (HOL) blocking

Wormhole Example

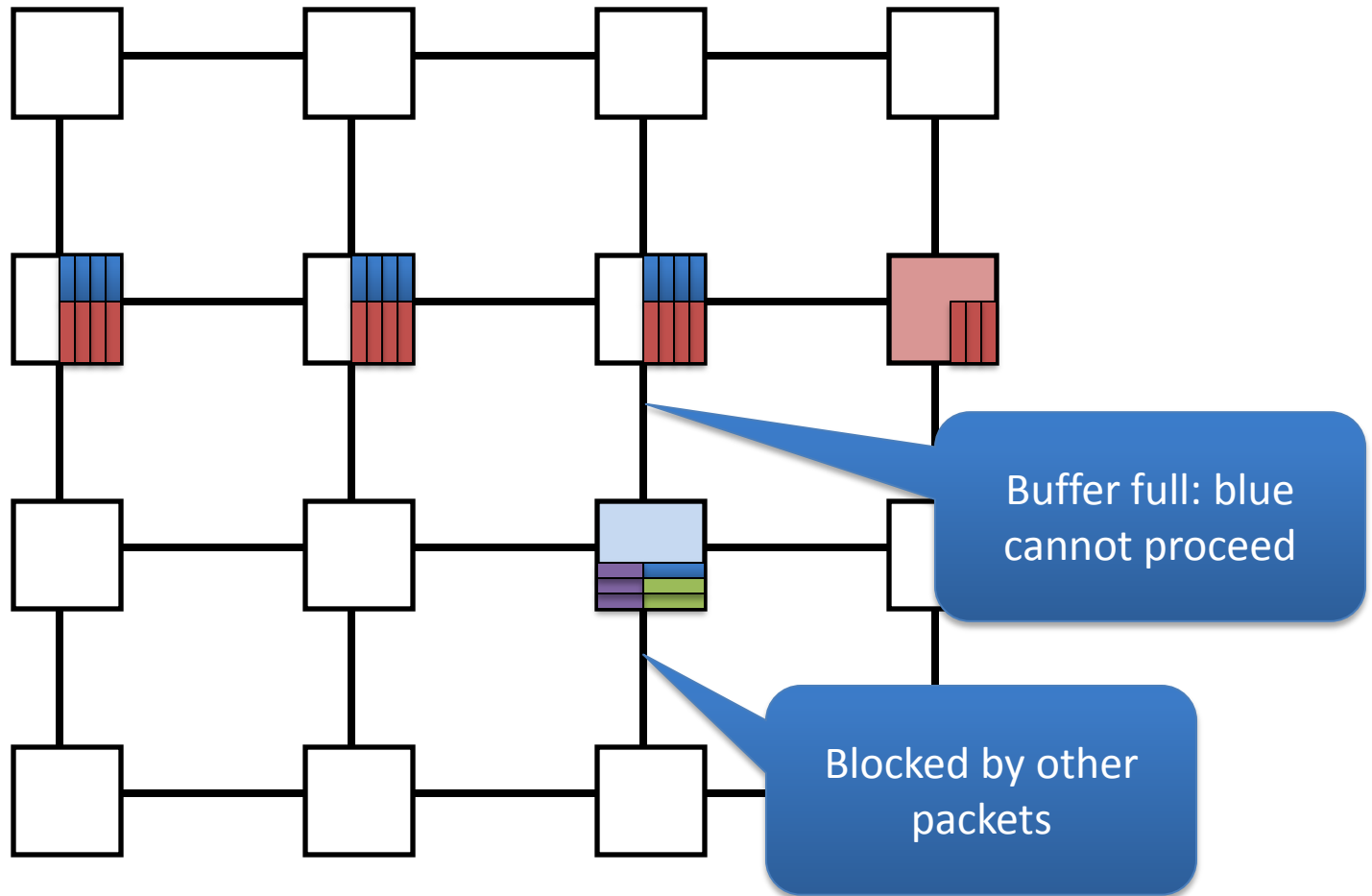


- 6 flit buffers/input port

Virtual Channel Flow Control

- Virtual channels used to combat HOL block in wormhole
- Virtual channels: multiple flit queues per input port
 - Share same physical link (channel)
- Link utilization improved
 - Flits on different VC can pass blocked packet

Virtual Channel Example



- 6 flit buffers/input port
- 3 flit buffers/VC

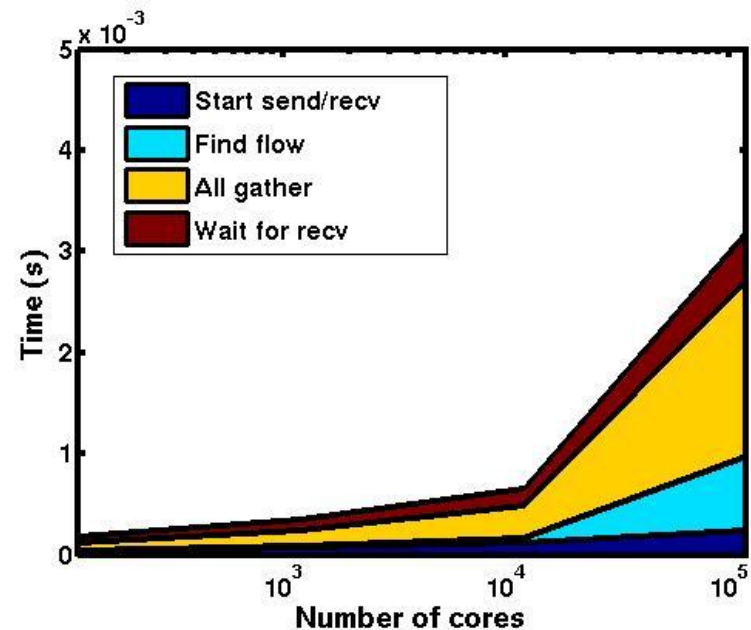
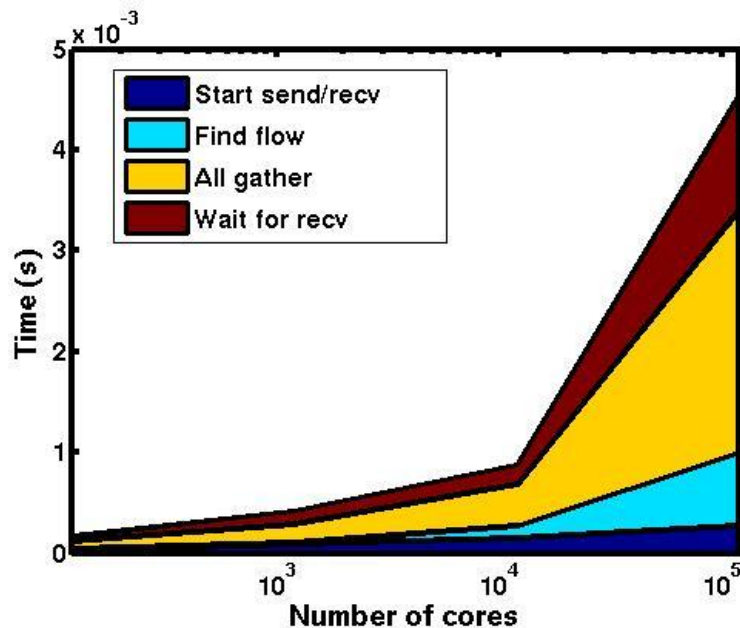
Why the need for Topology Mapping now ?

- Large-scale systems are built with low-dimensional network topologies
- E.g., 3D-Torus Jaguar (18k nodes), BG/P (64k nodes)
- Number of nodes grows ($\sim 100\text{k}$ -1M for Exascale)
- At this large scale, high chance of network congestion, hence advantages of hop count independence of wormhole routing are not applicable.
- Problem has been analysed for mapping Cartesian topologies [Yu'06, Bhatele'09, Krishna'11], arbitrary topologies [Hoefler'11]

The Mapping Problem

- **Definition:** Given a set of communicating parallel “entities”, map them on to physical processors to optimize communication
- **Goals:**
 - ✓ Minimize communication traffic and hence contention
 - ✓ Balance computational load (when $n > p$)
- **Case Study: Petascale Quantum Monte Carlo Application**

Task Assignment in Load Balancing

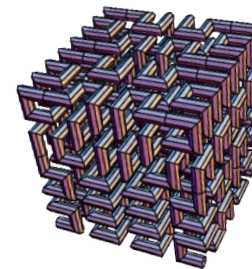


Time taken for different components of the new load balancing scheme with the default process ranks from MPI (left) and with our assignment (right)

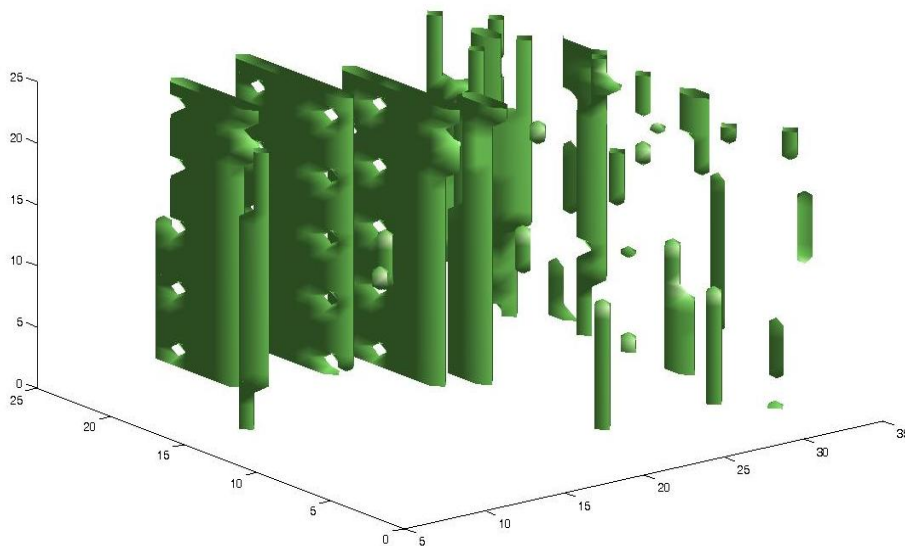
- Our assignment reduces send/rcv wait time by up to 60%
 - ✓ It reduces MPI_Allgather too by up to 30%

**Paper published in Elsevier Computer Physics Communications Journal
(Impact Factor: 3.268)**

Task-Node Affinity



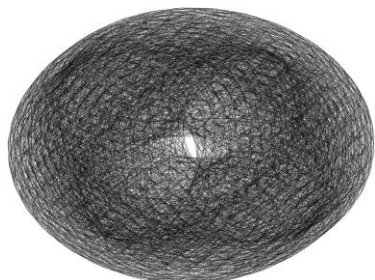
- Our task assignment for load balancing used a **3-D space filling curve**, assuming that the nodes are predominantly in a few cubic pieces of the machine
 - ✓ This assumption is not accurate
 - ✓ A more general solution will be useful



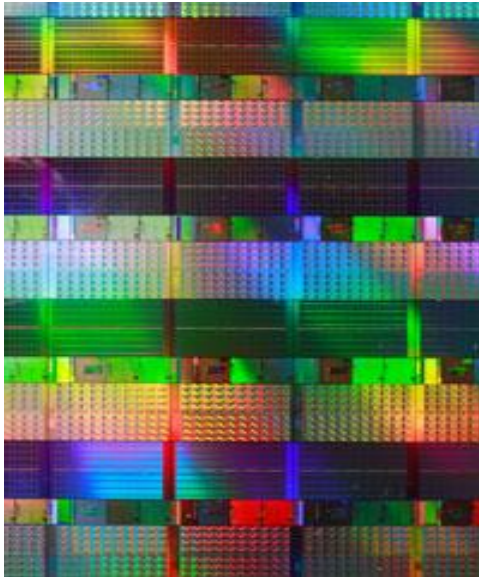
Nodes allocated for a run with 1K nodes on Jaguar

Cray XT5 (NCCS Jaguar)

18,688 nodes, SeaStar 2+
25x32x24 **3D-Torus**
network
2.595 PF, 532 TB/s
interconnect

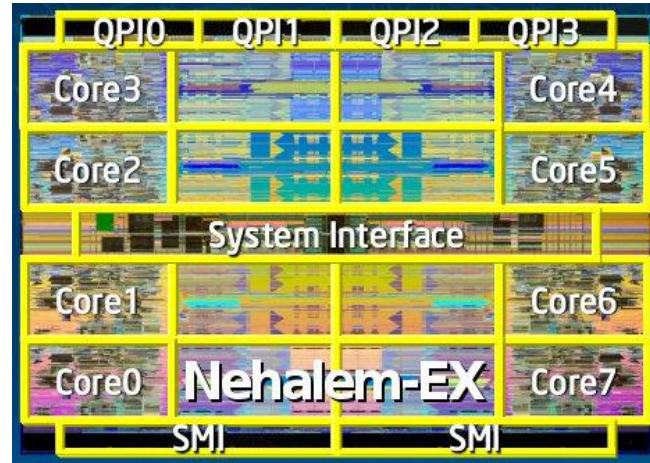


On-Chip Interconnect networks



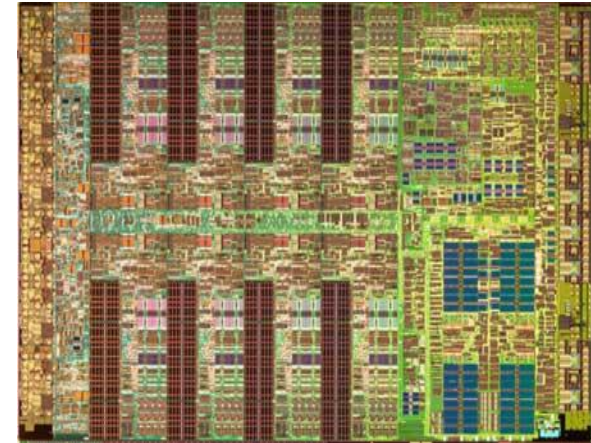
Intel Polaris

80-core prototype
2D Mesh



Intel Nehalem EX

Ring



IBM Cell BE

Ring

Sun Niagara

Crossbar

MIT Raw, TRIPs

2-D Mesh Topology

Cell BE Processor Architecture

- **Cell BE**

8 SPEs, 1 PPE

EIB

- **Inter SPE Communication:**

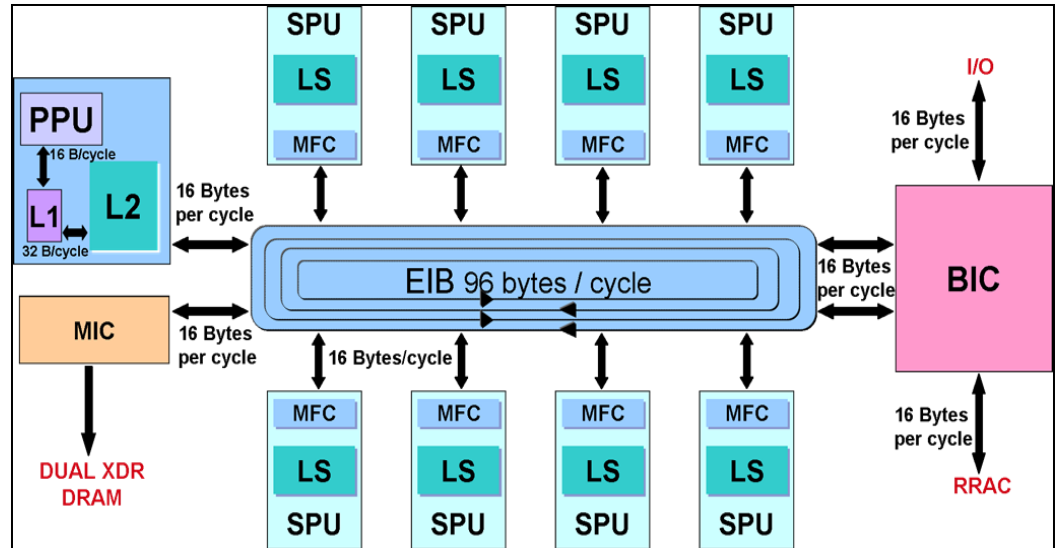
EIB theoretical

peak: 204.8 GB/s

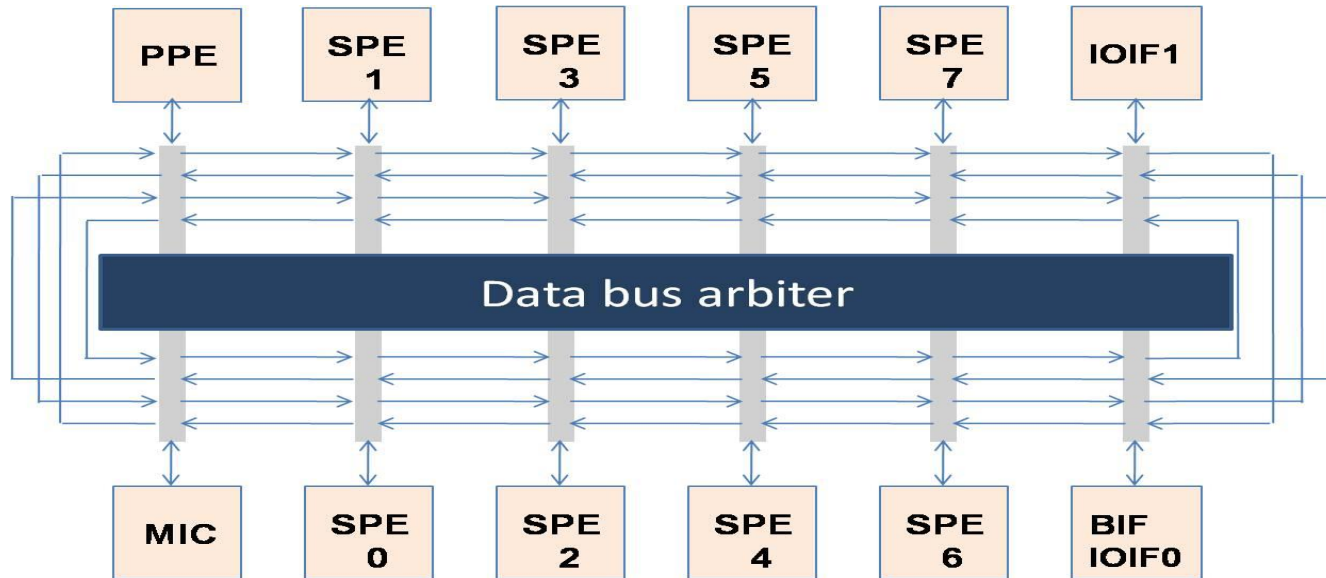
- **Memory Access:**

MIC 25.6 GB/s

- **Algorithm Design:** Advantageous if SPEs communicate directly over EIB, and have less main memory usage.

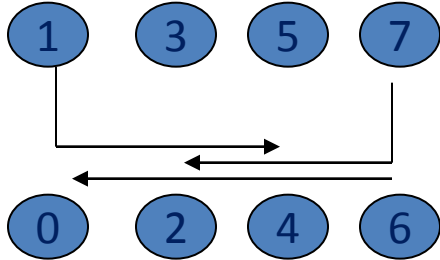


Cell BE Topology and Routing



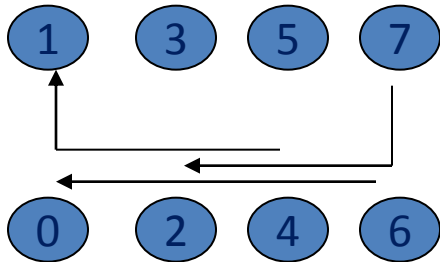
- **Topology:** Four unidirectional rings, two in each direction
- Theoretical peak network bandwidth is 204.8 GB/s
- Worst-case throughput of 50% or even less with adversarial traffic patterns
- **Routing:** Each ring supports 3 transfers when no path overlap
- Only shortest path routes are permitted

Inter-SPE Communication Bandwidth Analysis



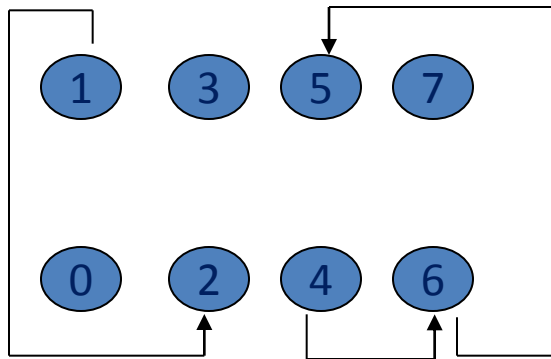
Three Comm. overlap
and not all in same direction

x06=[25.5969];
x27=[25.5969];
x41=[25.5642];



Three – Comm. overlap
and all in same direction

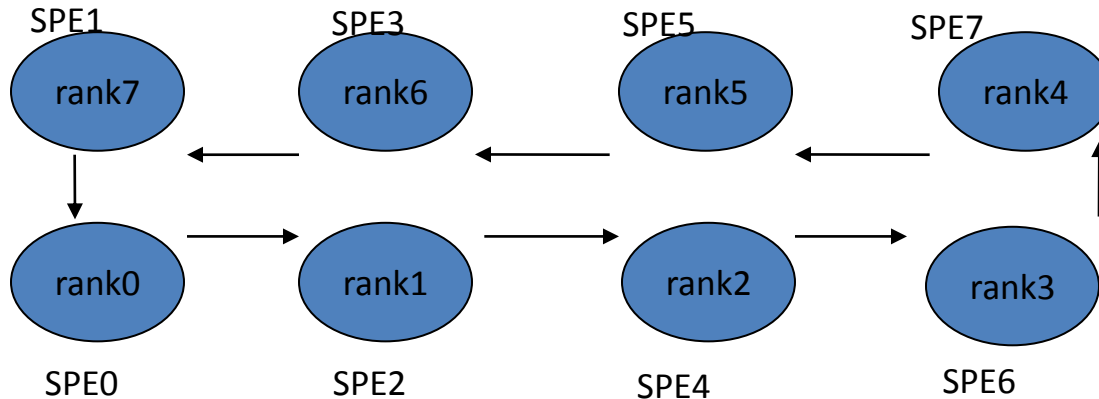
x06=[17.052];
x27=[25.5362];
x14=[17.0661];



Three non overlapping comm.
and in one direction and on the
same row.

x21=[24.54];
x64=[24.15];
x56=[24.3];

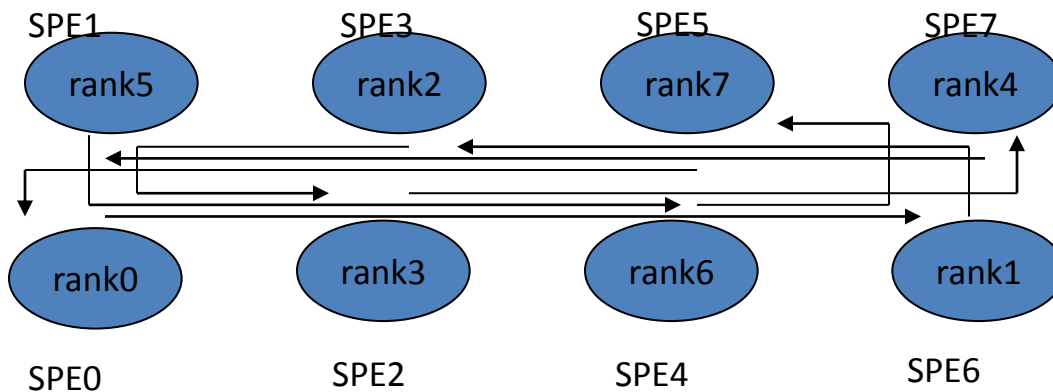
Performance of the Ring Pattern



Ring Mapping

MIN. Bandwidth
[14.61 GB/s];
AVG Bandwidth
[15.21 GB/s];

**NO CONGESTION,
LOAD IMBALANCE**

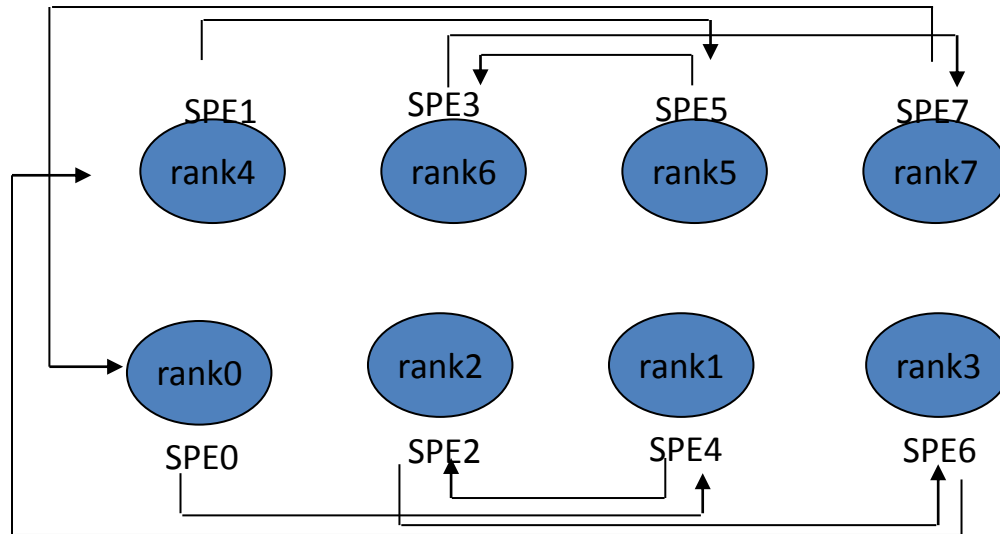


Overlap Mapping

MIN Bandwidth
[7.24 GB/s];
AVG Bandwidth
[8.8 GB/s];

**CONGESTION &
LOAD IMBALANCE**

Performance of the Ring Pattern



MIN. Bandwidth
[24.15 GB/s];
AVG Bandwidth
[24.33 GB/s];

**NO CONGESTION,
LOAD BALANCED.**

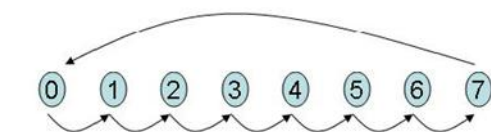
EvenOdd Mapping

Observations:

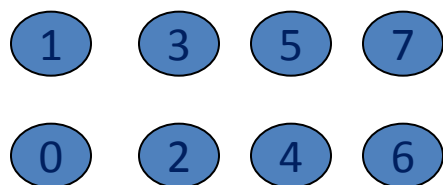
- Avoid overlapping paths for more than two messages in the same direction.
- Minimize the number of messages in any direction by balancing the load in both directions.
- Do not make any assumptions regarding the direction of transfer for messages that travel half-way across the EIB ring.

Performance of the Ring Pattern

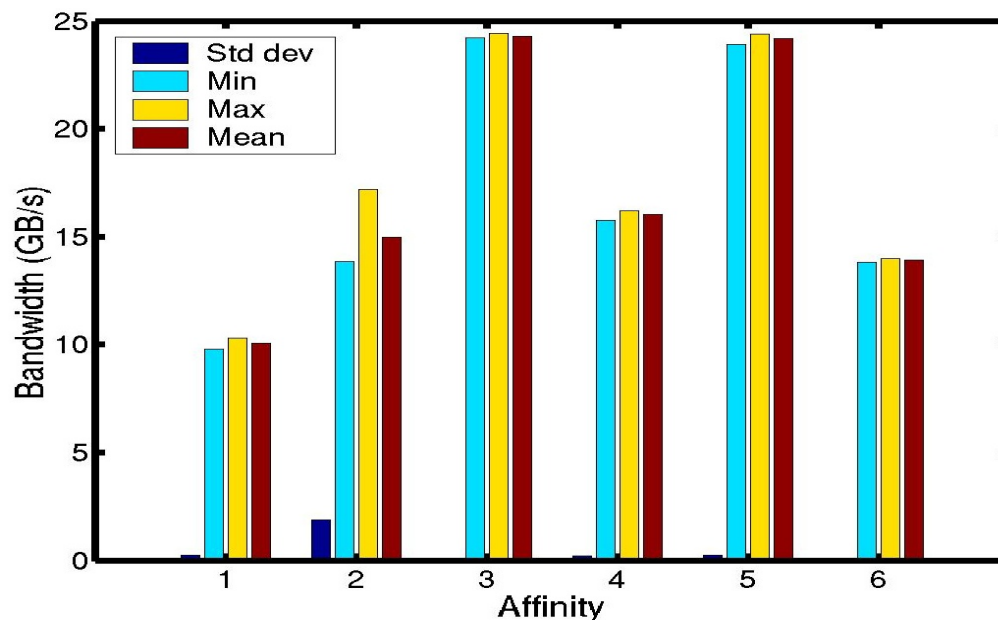
Affinity	(Physical ID, Thread Number) mapping
<i>Overlap</i>	{ (0, 0), (1, 7), (2, 2), (3, 5), (4, 4), (5, 3), (6, 6), (7, 1) }
<i>EvenOdd</i>	{ (0, 0), (1, 4), (2, 2), (3, 6), (4, 1), (5, 5), (6, 6), (7, 2) }
<i>Identity</i>	{ (0, 0), (1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6), (7, 7) }
<i>Leap2</i>	{ (0, 0), (1, 4), (2, 7), (3, 3), (4, 1), (5, 5), (6, 6), (7, 2) }
<i>Ring</i>	{ (0, 0), (1, 7), (2, 1), (3, 6), (4, 2), (5, 5), (6, 3), (7, 4) }



Ring Pattern

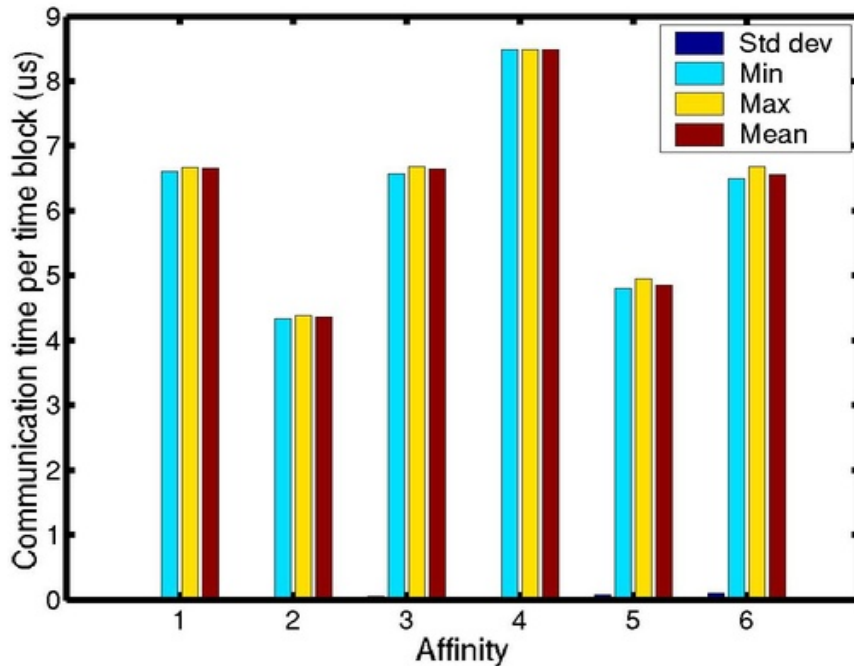


SPE Physical Layout

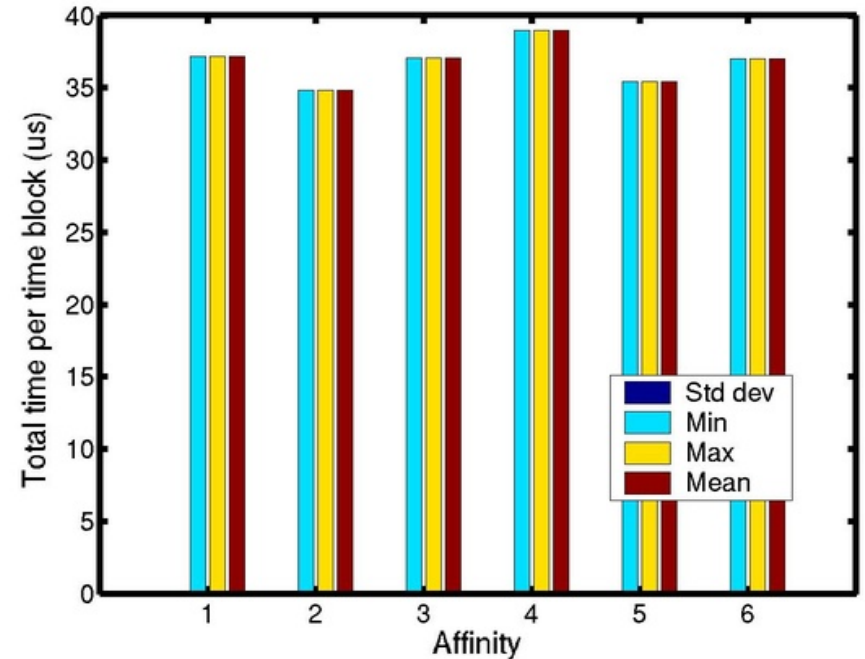


Affinities Tested: 1. Overlap 2. Default 3. EvenOdd 4. Identity 5. Leap2 6. Ring

Performance of Particle Transport Application



Communication Time



Total Application Time

Affinities Tested: 1. Identity 2. EvenOdd 3. Ring 4. Overlap 5. Leap2 6. Default

**A factor of 2 difference between
the best and worst affinities**

**10% between the best and
worst affinities**

Paper published in IEEE IPDPS 2009, PDSEC workshop;
PhD proposal accepted at IEEE TCPP PhD Forum